Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Thermoplastic pultrusion technology for industrial-scale production of recyclable continuous fiber-reinforced composites			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery-materials		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / railway vehicles	Body & lightweight technology	Fiber material	Fiber-reinforced composite material
Concept of Proposed Technology	• *Development of an **industrial thermoplastic pultrusion process** for continuous production of fiber-reinforced composite profiles* • *Continuous **melt impregnation of endless fibers** using thermoplastic matrices instead of conventional thermosets* • *Integration of **heating, impregnation, shaping, and cooling** into a stable, automated pultrusion process* • *Processing of **various reinforcement formats** (e.g. rovings, textiles, tapes, commingled yarns)* • *Enable production of **structural composite profiles** with constant cross-sections and high fiber volume content*			

	• *Focus on **process stability, reproducibility, and industrial scalability*** • *Compatibility with **recyclable thermoplastic material systems** and closed-loop manufacturing concepts* • *Basis for **downstream processes** such as thermal forming, over-molding, or joining* • *Applicable to **automotive, rail, construction, energy, aviation and industrial lightweight structures***
Rationale for Joint R&D Project	• *Thermoplastic pultrusion is a **high-potential manufacturing technology** but still limited in **industrial implementation and large-scale production*** • *Further development requires **close interaction between applied research and industrial manufacturing*** • *Research institute expertise is needed for **process physics, material–process interaction, and technology development*** • *Korean SME involvement is essential for **industrial scaling, production-oriented optimization, and commercialization*** • *Joint R&D enables **risk sharing** in developing and validating novel thermoplastic pultrusion process chains* • *Collaboration supports **technology transfer** from laboratory and pilot scale to **industrial production environments*** • *Korean SMEs provide **market access, application know-how, and production feedback** that cannot be generated by research institutes alone* • *Joint development allows **adaptation of the technology to Korean industrial standards, equipment, and supply chains*** • *International collaboration accelerates **time-to-market** and reduces development cycles* • *The project aligns with Korea's strategy for **advanced manufacturing, lightweight structures, and sustainable materials***
Key Words	1. Thermoplastic pultrusion 2. Fiber-reinforced composites 3. Lightweight design 4. Cost efficient manufacturing 5. Future mobility

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	This project focuses on optimizing and industrialization of the thermoplastic pultrusion process for endless fiber reinforced parts: Key Activities: 1) Heating, Tooling, Material Handling, and Conditioning: Development of precise control systems for material processing. 2) Reinforcing Materials: Investigation of diverse reinforcing materials such as rCF, rovings, textiles, tapes, and commingled yarns. 3) Injection Chamber Design: Investigation of various injection chamber designs to ensure optimal impregnation. 4) Generating Off-Axis Reinforcement: Continuous impregnation of off-axis reinforcement to achieve superior reinforcement properties. 5) Surface Optimization: Improving surface quality for both parts and tooling to enhance durability and performance. Concepts for Reducing Void Content: Addressing and minimizing voids to improve mechanical properties and process reliability.			
R&D Target Specifications	***Mechanical Performance*** - *Tensile strength: ≥ 600 MPa* - *Tensile modulus: ≥ 25 GPa* - *Flexural strength: ≥ 300 MPa* - *Flexural modulus: ≥ 25 GPa* - *Compressive strength: ≥ 180 MPa* - *Shear strength: ≥ 20 MPa* ***Material Quality*** - *Void volume content: ≤ 2 %* ***Productivity*** - *Production speed ≥ 1 m/min*			
R&D Plan	**Task 1: Process and Tooling Development** • *Design and optimization of heating, impregnation, and shaping concepts for thermoplastic pultrusion* • *Development of process-specific tooling including injection chamber, dies, and cooling concepts* • *Experimental evaluation of temperature profiles, pressure levels, and residence times* *Methodologies:* • *Experimental trials on industrial-scale pultrusion lines* • *Process parameter studies and design-of-experiments* • *Inline and offline process monitoring* ***Task 2: Material and Reinforcement Evaluation*** • *Investigation of thermoplastic matrix systems suitable for continuous pultrusion* • *Evaluation of reinforcement formats including rovings, textiles, tapes, and commingled yarns*			

	• *Optimization of fiber impregnation and wet-out behavior* *Methodologies:* • *Comparative material trials* • *Microscopy and void content analysis* • *Mechanical testing according to ASTM and ISO standards* ***Task 3: Process Stabilization and Quality Optimization*** • *Reduction of void content and surface defects* • *Improvement of dimensional stability and surface quality* • *Establishment of stable process windows for continuous operation* *Methodologies:* • *Long-duration pultrusion trials* • *Statistical evaluation of process stability* • *Surface and quality inspections* ***Task 4: Performance Validation*** • *Mechanical characterization of pultruded profiles* • *Validation of tensile, flexural, compressive, shear, fatigue, and impact properties* • *Correlation of process parameters and material performance* *Methodologies:* • *Standardized mechanical testing according to ASTM and ISO* • *Data-driven analysis of structure–property relationships* ***Task 5: Pilot-Scale Demonstration and Technology Transfer Preparation*** • *Pilot-scale production of representative composite profiles* • *Assessment of industrial production speed and process reliability* • *Preparation of guidelines for technology transfer to Korean SME* *Methodologies:* • *Pilot-scale pultrusion runs* • *Documentation of process parameters and best practices* • *Joint workshops and technical training* ***Expected Milestones and Timeline*** • *Year 1: Definition of process concepts and tooling designs, initial material screening and feasibility demonstration* • *Year 2: Optimization of process parameters and material combinations, achievement of target mechanical and quality specifications at pilot scale* • *Year 3: Stable high-speed pultrusion at or above 1 m/min, full validation of target specifications, preparation for industrial implementation at Korean SME partner*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IGCV has over a decade of experience in pultrusion and has successfully transferred multiple pultrusion processes into industrial applications (e.g. a seat-carrier structure for the automotive industry, https://www.igcv.fraunhofer.de/en/research/reference_projects/large_series_automotive_cfrp_structure_part.html) as well as a proven academic track record [1-15]. With two industrial-scale pultrusion lines and a versatile range of peripheral equipment (see **Fehler! Verweisquelle konnte nicht gefunden werden.**) - including extruders, braiders, and winders - the institute provides the technological foundation to further develop thermoplastic pultrusion as a scalable and industrially viable process. Key contributions: • **Process development & optimization:** Fraunhofer IGCV will lead the development and refinement of the thermoplastic pultrusion process, ensuring high process stability, repeatability, and material integrity to meet industrial requirements. • **Technology validation & scaling:** Through extensive testing and iterative improvements, Fraunhofer IGCV will advance the thermoplastic pultrusion process to a higher Technology Readiness Level, preparing it for industrial implementation. • **Process-specific tooling & equipment development:** Design and optimization of thermoplastic pultrusion-specific tools, fixtures, and auxiliary systems to enable seamless integration into industrial manufacturing environments. • **Material characterization & performance testing:** Fraunhofer IGCV will conduct mechanical testing, microscopy, and process analytics to assess key properties such as surface quality, void content, and mechanical performance, ensuring suitability for demanding applications. • **Training & knowledge transfer:** Fraunhofer IGCV will provide specialized training sessions and workshops to the Korean partner, ensuring that expertise in thermoplastic pultrusion is effectively transferred and implemented. • **Technology transfer & industrialization support:** Supporting the set-up and adaptation of the technology at the Korean partner's facility, ensuring a smooth transition from research to industrial application. Through these contributions, Fraunhofer IGCV will play a crucial role in developing, validating, and transferring thermoplastic pultrusion, driving industrial innovation, and strengthening the competitiveness of composite manufacturing in global markets.
Expected R&R of Korean SME Partner	The Korean SME(s) could play a key role in industrializing and commercializing the developed thermoplastic pultrusion process. Their primary focus could be on adapting the technology for large-scale production, developing market strategies, and ensuring the long-term economic viability of the process. Additionally, the SME(s) may explore complementary process developments and digital tools to enhance the technology's integration into industrial applications. Potential contributions and areas of focus: • **Industrialization & scaling:** The Korean SME(s) could support the adaptation and integration of the thermoplastic pultrusion process into an industrial production environment, ensuring process optimization for efficiency, cost-effectiveness, and quality standards. • **Development of a virtual process chain & simulation tools:** To enable a smooth transition to industrial production, the SME(s) could contribute to the development of

	a virtual process chain, including simulation and modeling of the thermoplastic pultrusion process. This would provide predictive analysis, process optimization, and a deeper understanding of material behavior, reducing development risks and accelerating implementation. • **Commercialization & market strategy:** A key responsibility could be developing a market entry strategy, identifying target industries and applications where the thermoplastic pultrusion process can provide a competitive advantage. Potential sectors include lightweight components for automotive, aerospace, hydrogen infrastructure, and construction, among others. • **Technology exploitation & IP utilization:** The SME(s) could lead efforts to explore patent protection, technology licensing, and potential partnerships to ensure successful commercialization of the developed process. • **Process chain optimization & potential follow-up processes:** Beyond pultrusion, the SME(s) could investigate additional downstream processes, such as injection molding, over-molding, or automated finishing steps, to create a fully integrated production chain and expand the range of potential applications. • **Production readiness & supply chain development:** Ensuring a reliable supply chain for materials, tooling, and equipment will be critical for commercialization. The SME(s) could assist in establishing partnerships with suppliers and manufacturing service providers. • **Demonstrator development & validation:** The SME(s) could contribute to the production, testing, and validation of demonstrator components, ensuring that the technology meets industry-specific mechanical, durability, and regulatory requirements. • **Strategic networking & industry engagement:** To drive adoption of the thermoplastic pultrusion process, the SME(s) could actively engage with potential customers, suppliers, and regulatory authorities through trade fairs, industry consortia, and certification bodies. Through these potential contributions, the Korean SME(s) could help bridge the gap between research and industrialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes • Establishment of a validated **industrial thermoplastic pultrusion process** for continuous fiber-reinforced composite profiles • Achievement of target mechanical properties, low void content, and stable high-speed production • Demonstration of reliable melt impregnation of endless fibers with thermoplastic matrices • Availability of a **scalable process chain suitable for industrial implementation** • Improved recyclability and material efficiency compared to thermoset-based pultrusion processes Commercialization and Industrial Impact • Readiness for **technology transfer** and implementation at a Korean SME partner • Basis for commercialization of thermoplastic pultruded composite profiles in multiple industries • **Expansion of Korean SME manufacturing capabilities** in advanced composite production • **Increased competitiveness of Korean SMEs** in global lightweight and advanced manufacturing markets Economic Impact • **Reduction of manufacturing costs** through continuous, automated, and low-scrap production • Creation of **new business opportunities** and revenue streams **for Korean SMEs** • Strengthening of domestic supply chains for thermoplastic composite materials and equipment Broader Impact • Contribution to **sustainable manufacturing** through recyclable thermoplastic composite solutions • Support of national strategies for advanced manufacturing and carbon reduction • **Acceleration of international collaboration** between Korean industry and applied research institutes • Enhancement of global technology leadership in thermoplastic composite manufacturing

Impressions of »Pultrusion« at Fraunhofer IGCV

Figure 1: Impressions of recent pultrusion projects at Fraunhofer IGCV

Figure 2: Pultrusion machine Pultrex PX500 – 10T at Fraunhofer IGCV (one of two industrial scale machines)

6. References

1. Bauer, C.: »Analyzing the influence of process parameters on antistatic pipes produced by braiding pultrusion«; SAMPE Europe, 2025
2. Schwab, S.; Rieger, M.:»Thermoplastic pultrusion paves the way to mass application«, NAPC 2023
3. Rieger, M.; Du, T.; Senz, A.; Sauer, M.: »Pultrudates for loadpath reinforced injection molded parts – Analysis of thermoplastic melt pultruded rods«, SAMPE Europe, 2023
4. Frücht, M.; Senz, A.; Sydow, S.; Frank, J.; Hochmann, A.: »Sustainable Pultruded Sandwich Profiles with Mycelium Core«, Polymers, Vol.15, Issue 15
5. Strauß, S.; Wilhelm, F.; Senz, A.; Engelen, H.; Boyson, S.; Rilli, N.; Celik, A.; Ratka, M.; Bonten, C.: »Experimental and simulative analysis of the pressure development in a closed injection pultrusion process with multiple chamber geometries«, Polymers, 2023, 10.3390/polym15061544
6. Strauß, S.; Boysen, S.; Senz, A.; Wilhelm, F.; Rilli, N.: »Analysis of the mechanical composite properties of ii-chamber variations in the closed injection pultrusion process«, esaform, 2021, 10.25518/esaform21.970
7. Wilhelm, F.; Strauß, S.; Weigant R.; Drechsler K.: »Effect of Power Ultrasonic on the Expansion of Fiber Strands«, Journal of Composites Science, 4(2) 50, 2020; 10.3390/jcs4020050
8. Strauß, S.; Wilhelm, F.: »Development of a flexible injection and impregnation chamber for pultrusion of high reactive resins«, Procedia Manufacturing, Vol. 47, 2020; 10.1016/j.promfg.2020.04.294
9. Wilhelm, F.; Strauß, S.; Kronseder, M.: »Influence of power ultrasonic on the viscosity of epoxy resin«, Results in Materials, 2020; 10.1016/j.rinma.2020.100129
10. Strauß, S.: »Pultrusion of Polyurethane and Applications« 15th World Pultrusion Conference, Antwerp (Belgium), Feb. 2020

11. Mentizi, S.; Hecking, A.; Strauß, S.; Achten, D.; Guiteras, M. A.: »Aliphatic polyurethan matrix for sustainable UV-resistant composites«, Revista de la Asociacion Espanola de Materiales Compuestos, Vol. 3 No. 4, 2019; 10.23967/r.matcomp.2022.05.007
12. Strauß, S.; Meisenheimer R.; Hecking A.; Senz A.; Wilhelm, F.: »Manufacturing of predominatly bio-based composite profiles with pultrusion« 22nd International Conference on Composite Materials (ICCM22), Melbourne (Australia), Aug. 2019
13. Strauß, S.: »Flexibility in Polyurethan Injection Molding«, North American Pultrusion Conference, Rosemont (United States), Apr. 2019
14. Strauß, S.; Senz, A.; Ellinger, J.: »Comparison of the Processing of Epoxy Resins in Pultrusion with Open Bath Impregnation and Closed-Injection Pultrusion«, Journal of Composites Science, 3(3), 87, 2019; 10.3390/jcs3030087
15. Bezerra, R.; Henning, F.: »Modelling and simulation of the pultrusion process with closed injection and impregnation molds«, 17th European Conference on Composite Materials (ECCM17), 2016